$*$ & $**$ imply $P(\tilde{w}_1, \dots, \tilde{w}_{n+1}) = (\tilde{w}_1, \dots, \tilde{w}_{n+1})$. Thus $(\tilde{w}_1, \dots, \tilde{w}_{n+1}) \in W$.

Step 10 Since $\tilde{w}_1, \dots, \tilde{w}_{n+1} \in V$ are arbitrary, Step 9 indeed implies $W = V^{\oplus(n+1)}$.

But this contradicts Step 4 : $V^{\oplus(n+1)} = W \oplus U$ and $U \neq \{0\}$. ▤

Corollary 4.1.6 Let A and V be as in Theorem 4.1.5. For every subspace $X \subseteq V$ such that $\dim X < \infty$, and for every linear transformation $T : X \longrightarrow V$, there exists an element $a \in A$ such that $a|_X = T$.

Corollary 4.1.7 (Burnside's Theorem) Let V be a vector space and let $A \subseteq \mathrm{End}(V)$ be a unital subalgebra. Assume that $\dim V < \infty$ and V is an irreducible A-module. Then $A = \mathrm{End}(V)$.

Complete reducibility In this section A is any associative algebra.

Def 4.1.8 A finite dimensional A-module V is called completely reducible if for every A-submodule $W \subseteq V$ there exists an A-submodule $U \subseteq V$ such that

$$V = W \oplus U .$$

Lemma 4.1.9 Submodules and quotients of completely reducible A-modules are completely reducible.

Proof $W_1 \subseteq W \subseteq V \Rightarrow V = W_1 \oplus U$ Now set $U_1 := W \cap U$. Then $W = U_1 \oplus W_1$.

Similarly, $W \subseteq V$ and $W/W \subseteq V/W \Rightarrow$ Choose $U \subseteq V$ such that $V = W_1 + U$.

Then $V/W \simeq W/W \oplus (U + W/W)$

Proposition 4.1.11 Let V be an A-module. Assume that $\dim V < \infty$. Then the following are equivalent :

(a) V is completely reducible

(b) $V = W_1 \oplus \cdots \oplus W_r$ where W_i's are irreducible A-modules.

(c) $V = V_1 + \cdots + V_s$ where $V_1, \dots, V_s$ are irreducible A-submodules of V.

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Proof (a) $\Rightarrow$ (b) : Induction on $\dim V$

(b) $\Rightarrow$ (c) : trivial.

(c) $\Rightarrow$ (a) : Fix $W \subset V$. Let $\mathcal{C}$ be the family of submodules $M \subseteq V$ that satisfy the following conditions:

(i) $M = \sum_{\alpha \in I} M_\alpha$ for a family of simple submodules $M_\alpha \subset V$, $\alpha \in I$.

(ii) $M \cap W = \{0\}$.

Step 1 Choose $\tilde{M} \in \mathcal{C}$ such that $\dim \tilde{M}$ is maximum possible. We claim that $V = W \oplus \tilde{M}$.

Step 2 $\tilde{M} \cap W = \{0\}$ by definition of $\tilde{M}$.

Step 3 Assume $\tilde{M} + W \neq V$. Since V is a sum of simple submodules, there exists a simple submodule $M_0 \subset V$ such that $M_0 \not\subseteq \tilde{M} + W$. Since M_0 is irreducible, we have $M_0 \cap (\tilde{M} + W) = \{0\}$

Step 4 Set $\tilde{\tilde{M}} := \tilde{M} + M_0$. Clearly $\dim \tilde{\tilde{M}} > \dim M$, hence $\tilde{\tilde{M}} \cap W \neq \{0\}$.

Step 5 $x \in \tilde{\tilde{M}} \cap W \Rightarrow x = \tilde{x} + x_0$ ($\tilde{x} \in \tilde{M}$, $x_0 \in M_0$) $\Rightarrow x_0 = x - \tilde{x} \in \tilde{M} + W \Rightarrow x_0 = 0$

($x \in W$, $x_0 \in M_0$)

$\Rightarrow x = \tilde{x} \in \tilde{M} \Rightarrow x \in \tilde{M} \cap W = \{0\} \Rightarrow x = 0$ ∎

Corollary (Prop 4.1.11) Let (ρ, V) be a completely reducible A-module. Assume that $\dim V < \infty$. Also, assume that $V = \bigoplus_{i=1}^{d} V_i$, where the V_i's are all isomorphic to a single A-module W. Then every A-submodule of V is a direct sum of modules isomorphic to W.

Proof Fix a submodule $M \subseteq V$. By Prop. 4.1.11, $M = \bigoplus_{i=1}^{d} M_i$ where the M_i's are simple.

Step 1 Assume that $\exists_{1 \leq j \leq d} : M_j \neq W$. Then we consider the maps:

$$p_{j,i} : M_j \longrightarrow M \longrightarrow V \longrightarrow V_i \qquad 1 \leq i \leq d$$

Step 2 $p_{j,i}$ is an A-module homomorphism. Hence $p_{j,i} = 0$.

Step 3 It follows that $M_j = \{0\}$. Contradiction.

Corollary 4.1.12 Let (ρ, V) and (σ, W) be completely reducible A-modules. Then

$(\rho \oplus \sigma, V \oplus W)$ is also completely reducible.

Proof Trivial from Prop. 4.1.11.

Def Given a vector space V and a set $S \subseteq \mathrm{End}(V)$, we define the **Commutant** of S by $\mathrm{Comm}(S) := \{ x \in \mathrm{End}(V) : xs = sx \text{ for all } s \in S \}$.

Remark $\mathrm{Comm}(S)$ is always a unital subalgebra of $\mathrm{End}(S)$.